

Signals, Sampling and Filtering

Course notes - Department of Electrical Engineering

1. Signals and systems

A signal carries information as a function of time or space; a system transforms one signal into another. Linear time-invariant systems are the tractable case, fully described by their impulse response.

The output of an LTI system is the convolution of the input with the impulse response, which in the frequency domain becomes simple multiplication. That equivalence is why frequency-domain analysis dominates.

2. Fourier analysis

The Fourier series decomposes a periodic signal into harmonically related sinusoids. The Fourier transform extends this to aperiodic signals, mapping a time-domain signal to a continuous spectrum.

Duality runs throughout: narrow in time means broad in frequency. A perfect impulse contains all frequencies, and a pure sinusoid must last forever.

3. Sampling

Sampling multiplies a continuous signal by an impulse train, which replicates its spectrum at multiples of the sampling rate. If those replicas overlap, the original cannot be recovered.

The Nyquist-Shannon theorem requires a sampling rate above twice the highest frequency present. Aliasing folds higher frequencies down onto lower ones irreversibly, so an analogue anti-aliasing filter must precede the converter.

4. The discrete Fourier transform

The DFT maps N samples to N frequency bins and implicitly assumes the block repeats periodically. A signal that does not complete a whole number of cycles in the window produces spectral leakage, which windowing functions reduce at the cost of resolution.

The fast Fourier transform computes the DFT in $O(N \log N)$ rather than $O(N^2)$ by recursively exploiting symmetry. Its existence is what made digital signal processing practical.

5. Digital filters

Finite impulse response filters have no feedback, are always stable, and can be given exactly linear phase, which matters wherever waveform shape must be preserved.

Infinite impulse response filters use feedback to achieve a given magnitude response with far fewer coefficients, at the cost of possible instability and non-linear phase. Placing poles inside the unit circle is the stability requirement.